

MIP_PROMPT_LIBRARY_V2_PART5_TESTING_AND_QUALITY.

Mainframe Intelligence Platform

Enterprise Prompt Library V2

Part 5 - Testing, Validation & Quality Engineering

Version: 2.0

Status: Approved

Purpose

Trust is the product.

Metadata without validation is dangerous.

Graphs without verification are misleading.

Recommendations without explainability are unusable.

This section defines prompts used to validate, test, benchmark, certify, and continuously improve MIP.

Quality Engineering Principles

Verify Everything

Every extracted fact must be verifiable.

Explain Everything

Every recommendation must have evidence.

Fail Transparently

Unknowns are better than incorrect conclusions.

Measure Everything

Every parser, graph, and reasoning engine must produce metrics.

Continuous Validation

Validation is not a one-time event.

Prompt 21

Parser Validation Framework

File

prompts/testing/parser_validation.md

Purpose

Evaluate parser accuracy and completeness.

Context

Parsers are the source of truth for metadata generation.

Errors propagate into every downstream capability.

Inputs

- Parser Output
- Source Code
- Metadata Models

Instructions

Validate:

- Missing entities
- Incorrect entities
- Duplicate entities
- Invalid relationships
- Unsupported syntax

Calculate:

- Precision
- Recall
- Coverage

Expected Output

Validation Summary

Accuracy Metrics

Coverage Metrics

Failure Analysis

Recommended Fixes

Constraints

Every finding must reference source evidence.

Success Criteria

Parser quality becomes measurable and repeatable.

Prompt 22

Metadata Integrity Validation

File

prompts/testing/metadata_integrity.md

Purpose

Verify metadata consistency across the platform.

Context

Metadata becomes the canonical system representation.

Inputs

Metadata Repository

Instructions

Validate:

- Required attributes
- Missing references
- Orphan entities

- Circular references
- Duplicate entities

Generate integrity report.

Expected Output

Integrity Score

Entity Health Report

Relationship Health Report

Remediation Recommendations

Success Criteria

Metadata repository becomes self-validating.

Prompt 23

Graph Consistency Validation

File

prompts/testing/graph_validation.md

Purpose

Validate knowledge graph correctness.

Context

The graph powers impact analysis and modernization reasoning.

Inputs

Knowledge Graph

Metadata Repository

Instructions

Validate:

- Missing edges
- Invalid edges
- Broken paths

- Unreachable nodes
- Circular dependencies

Generate graph health report.

Expected Output

Graph Integrity Report

Traversal Validation

Dependency Validation

Risk Assessment

Success Criteria

Graph accuracy is measurable.

Prompt 24

End-to-End Pipeline Validation

File

prompts/testing/e2e_validation.md

Purpose

Validate complete MIP execution flow.

Context

The pipeline must operate consistently from discovery through reasoning.

Inputs

Repository

Instructions

Execute:

Repository

↓

Discovery

↓

Parsing

↓

Metadata

↓

Graph

↓

Reasoning

Validate every stage.

Expected Output

Pipeline Report

Stage Metrics

Failure Analysis

Confidence Score

Success Criteria

Entire platform becomes testable.

Prompt 25

Performance Benchmarking

File

prompts/testing/performance_benchmark.md

Purpose

Measure platform scalability.

Context

Enterprise repositories can contain:

- 50,000+ programs
- Millions of relationships

Inputs

Large Repository

Instructions

Measure:

- Parse Time
- Graph Build Time
- Query Time
- Memory Usage
- Storage Usage

Generate benchmark report.

Expected Output

Performance Dashboard

Scalability Metrics

Bottleneck Analysis

Optimization Recommendations

Success Criteria

Platform performance becomes predictable.

Prompt 26

Explainability Validation

File

prompts/testing/explainability_validation.md

Purpose

Ensure recommendations are explainable.

Context

Architects must trust MIP outputs.

Inputs

Reasoning Results

Recommendations

Instructions

For every recommendation:

Identify:

- Source Evidence
- Metadata Evidence
- Graph Evidence
- Reasoning Path

Generate explainability report.

Expected Output

Evidence Chain

Recommendation Trace

Confidence Score

Success Criteria

Every recommendation is auditable.

Prompt 27

Change Impact Validation

File

prompts/testing/change_impact_validation.md

Purpose

Verify impact analysis correctness.

Context

Impact analysis is one of MIP's highest-value capabilities.

Inputs

Target Component

Knowledge Graph

Instructions

Validate:

- Upstream dependencies
- Downstream dependencies
- Data impacts
- Batch impacts
- Online impacts

Compare results with graph traversal.

Expected Output

Impact Validation Report

Coverage Report

Risk Analysis

Success Criteria

Impact predictions are reliable.

Prompt 28

Data Lineage Validation

File

prompts/testing/data_lineage_validation.md

Purpose

Verify lineage accuracy.

Context

Lineage drives compliance and modernization planning.

Inputs

Lineage Graph

Metadata Repository

Instructions

Validate:

- Source systems
- Transformations
- Intermediate datasets
- Target systems

Generate lineage confidence report.

Expected Output

Lineage Health Report

Coverage Analysis

Gap Analysis

Success Criteria

End-to-end lineage is trustworthy.

Prompt 29

Knowledge Coverage Assessment

File

prompts/testing/knowledge_coverage.md

Purpose

Measure how much of the system MIP understands.

Context

Modernization decisions require confidence.

Inputs

Repository

Metadata

Graph

Instructions

Calculate:

- Repository Coverage
- Program Coverage
- Job Coverage
- Dataset Coverage
- Relationship Coverage

Generate coverage dashboard.

Expected Output

Coverage Metrics

Coverage Heat Map

Blind Spot Analysis

Success Criteria

Unknown areas become visible.

Prompt 30

Modernization Readiness Assessment

File

prompts/testing/modernization_readiness.md

Purpose

Determine whether the repository is ready for modernization planning.

Context

Many organizations attempt modernization too early.

Inputs

Repository

Metadata

Graph

Validation Results

Instructions

Assess:

- Metadata Quality
- Graph Quality
- Dependency Completeness
- Lineage Completeness
- Impact Analysis Reliability

Generate readiness score.

Expected Output

Modernization Readiness Report

Risk Matrix

Recommended Next Actions

Executive Summary

Success Criteria

Leadership receives objective modernization readiness metrics.

Enterprise Quality Gates

MIP should not proceed to modernization recommendations unless:

✓ Repository Coverage > 90%

✓ Metadata Integrity > 95%

✓ Graph Integrity > 95%

✓ Explainability Coverage > 95%

✓ Lineage Coverage > 85%

✓ Impact Analysis Confidence > 90%

Completion Criteria

MIP must now be able to answer:

- Is the metadata trustworthy?
- Is the graph accurate?
- Are recommendations explainable?
- Is the repository sufficiently understood?
- Is modernization safe to begin?

Only after these answers are positive should modernization planning begin.

End of Part 5.